Originality Statement

I hereby state that the code in this project was specifically written for the CP467 course at Wilfrid Laurier University, taught by Zia Ud Din. All code in this assignment was written by the student David Hu, unless stated otherwise below.

Brute force matching with L2, under match_descriptors(), taken from

https://docs.opencv.org/3.3.1/dc/dc3/tutorial_py_matcher.html

Method

For this project, the feature detection and description was done by SIFT, and I used OpenCVs brute force matcher. The chosen ratio_test and match_threshold values of 0.6 and 26 were reached after testing other values. Higher threshold values ended up not classifying as many objects correctly, while lower ones caused mismatches to appear. Similarly, the original Lowe ratio of 0.75 was too loose, and resulted in false matches, while lower final values ended up not properly detecting as many objects.

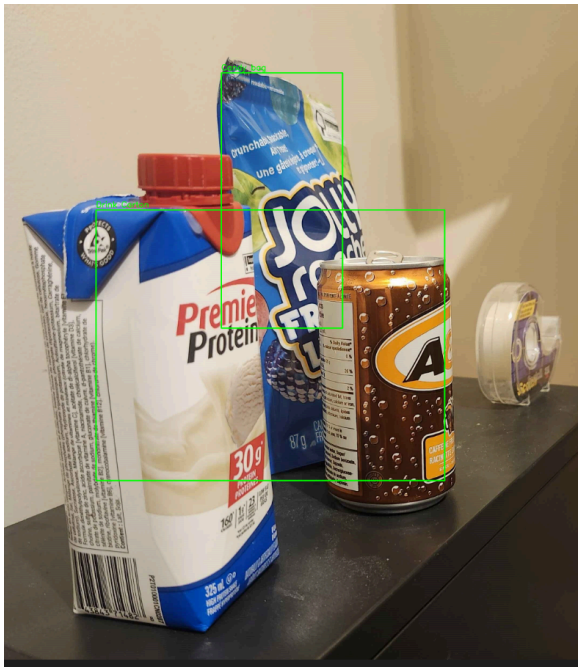
Results

After optimizations the program produced the following results:

Scene	O1	O2	O3	O4	Predicted	Ground truth
S1_front	593	274	34	24	O1, O2, O3	O1, O2
S1_left	45	52	19	15	O1, O2	O1, O2
S1_right	10	9	14	17	none	O1, O2
S2_front	239	39	299	133	O1, O2, O3, O4	O1, O3, O4
S2_left	37	25	67	10	O1, O3	O1, O3, O4
S2_right	28	14	105	28	O1, O3, O4	O1, O3, O4

Metric	Result
Precision	85.71%
Recall	80.00%
F1-score	82.76%
Accuracy	79.17%





S2_front and S2_left border box images. Note borders do not necessarily cover the object outline, only the matching features

Challenges

One challenge I encountered while working on this project was that SIFT sometimes produced false matches between an object image and unrelated parts of a scene, especially since the images were taken on a cluttered surface with background features such as curtains and wires. This initially caused some absent objects to exceed the detection threshold and be classified as present. I fixed this by cropping the images to get rid of as much background noise as possible and fine tuning my ratio and threshold values to 0.60 and 26 to reduce ambiguous matches while still keeping enough valid matches to detect the real objects.